KINGSTON UNIVERSITY LONDON

COURSEWORK ASSIGNMENT 2

Under the module

Data Analysis and Visualisation (CI7330)

M.Sc. Data Science

Submitted By

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Question 1: Red Wine Quality Analysis

Introduction

The increasing sales of red wine sales has led both producers and authorities to consider automated methods for quality assessment that can consistently produce expert human ratings. The present study scrutinizes the extent to which 11 physicochemical lab measurements can predict the wine quality scores given by the experts in the wine industry. It takes the dataset of 1599 red wine samples and starts with an exploration of the data features and variable relationships. It then employs multiple linear regression and subsequently performs manual backward elimination to find the simplest model that can be relied upon for making predictions. In the end, the model's validity and prediction power are assessed.

1. Initial Data Exploration

The dataset has **1599 complete cases** (i.e., no missing values). Basic statistics indicate that the quality variable is normally distributed (**mean 6.13, SD 0.86**), with the majority of the values falling within the range of 5 to 7. Alcohol, sulphates, and citric acid concentrations are the highest in wines of good quality, whereas those of volatile acidity, chlorides, and total SO□ are the lowest.

Key variables used:

- **fixacid:** Fixed acidity score
- **volacid:** Volatile acidity score
- **citric:** Citric acid content
- **residsugar:** Residual sugar content
- **chlorides:** Chlorides by volume
- **freeSO2:** Free sulphur dioxide content
- **totalSO2:** Total sulphur dioxide content
- **density:** Density score
- **pH:** pH level
- **sulphates:** Sulphates level
- **alcohol:** Alcohol content (vol)
- **Quality:** Wine quality based on human expert ratings(y=Outcome)

2. Exploratory Data Analysis

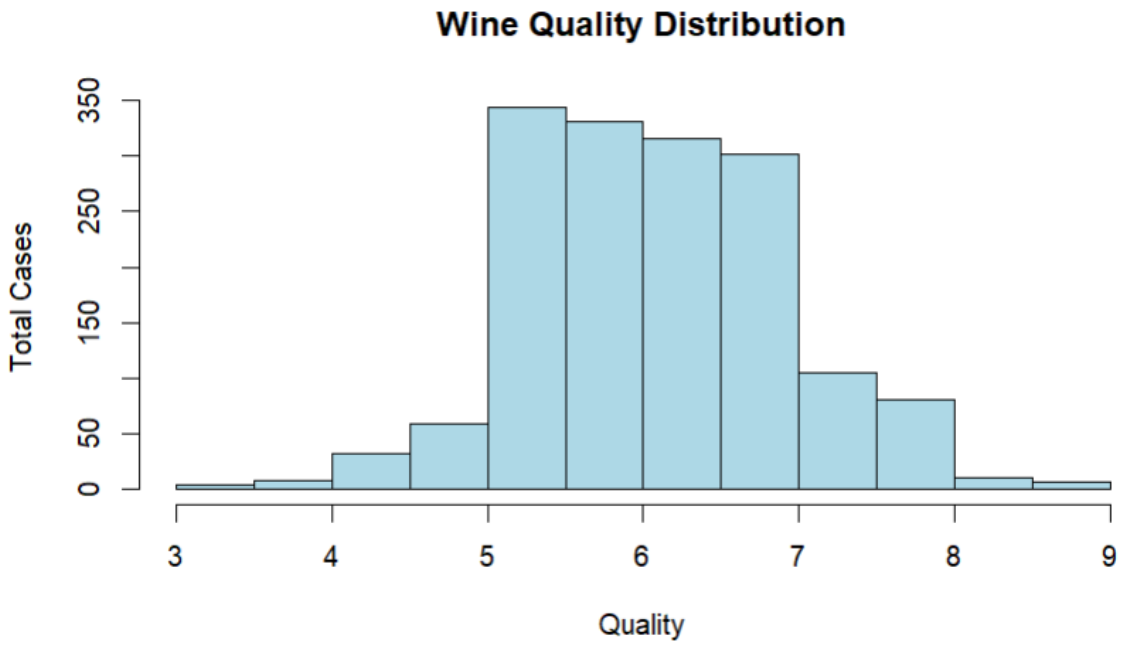


Figure 3.1: Wine Quality histogram

It can be seen from the histogram that the distribution of wine quality ratings is approximately normal, with the mean at 6 and the majority of the ratings between 5 and 7. Scoring below 4 or above 8 is very rare for wines.

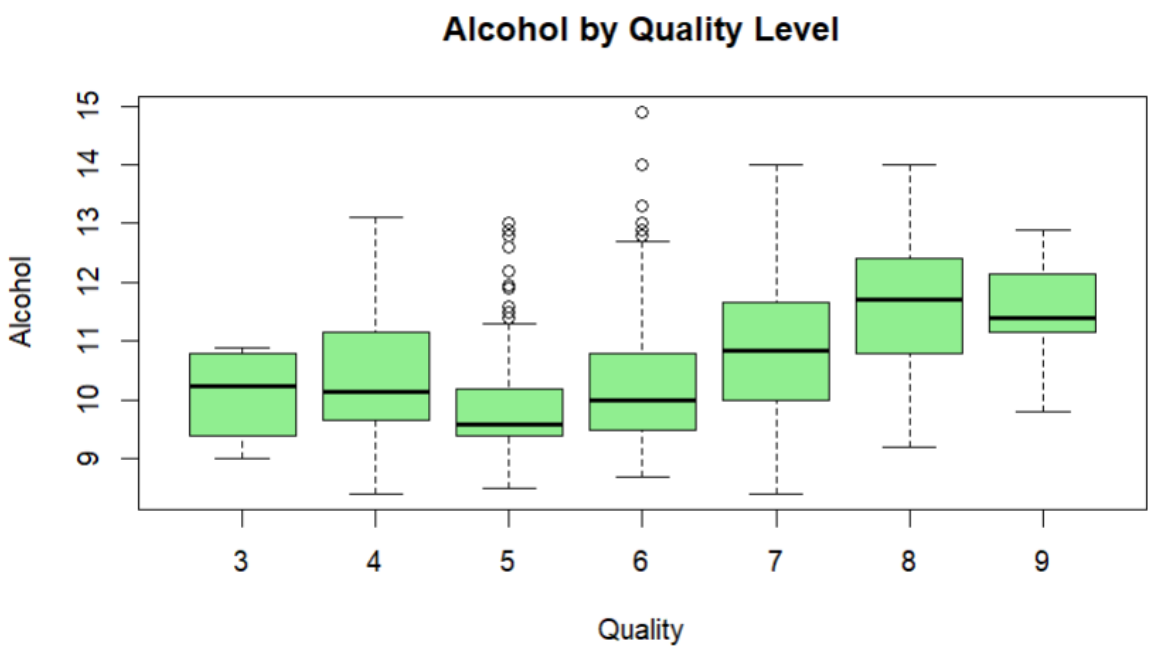


Figure 3.2: Alcohol boxplot at different quality levels

The boxplot depicts a strong positive correlation between quality and alcohol content quite clearly; the increase in median alcohol from about 9.5% in the case of lower-quality wines to over 11% in that of the higher-quality wines is quite noticeable.

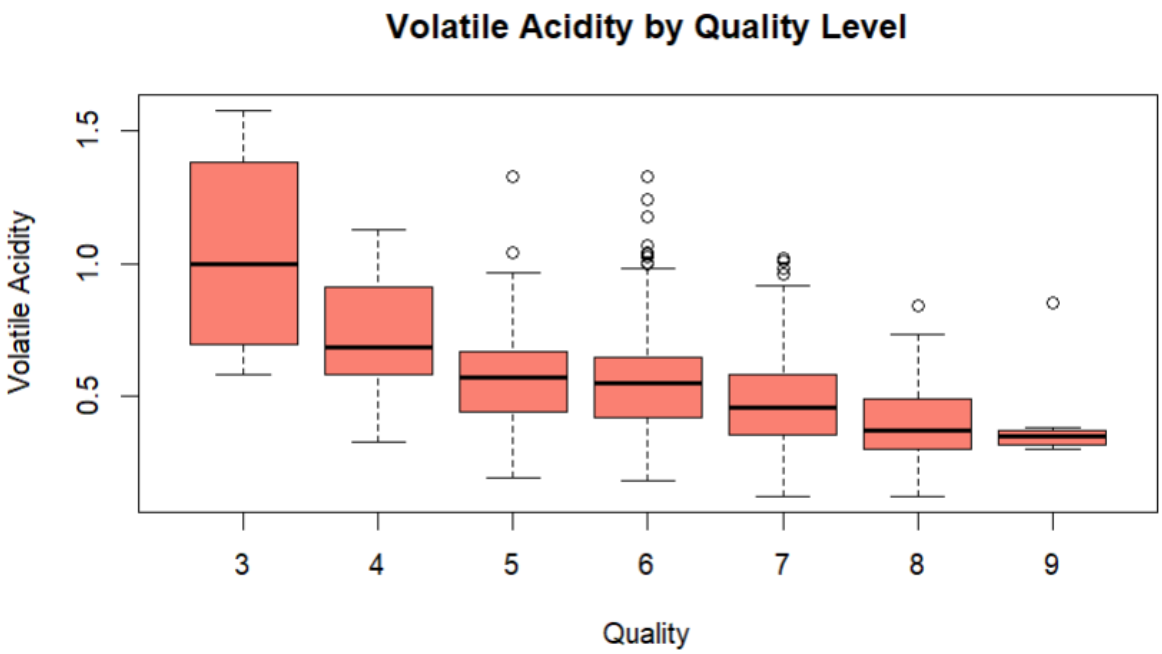


Figure 3.3: Volatile Acidity by Quality Level

The boxplot indicates a strong negative correlation: the lower the quality of the wine, the higher its volatile acidity, with the average values coming down gradually from about 0.7 in low-quality wines to around 0.4 in high-quality ones.

Correlation analysis shows:

- Strongest positive correlation: alcohol ($r \approx 0.45$)
- Strongest negative correlation: volatile acidity ($r \approx -0.36$)
- Moderate positive: sulphates ($r \approx 0.24$), citric ($r \approx 0.20$)
- Moderate negative: chlorides, total SO□, density

Slight multicollinearity is present (fixed acidity ↔ citric acid ≈ 0.67; freeSO2 ↔ totalSO2 ≈ 0.66), which is usual and will be dealt with during model selection.

The data is proper and appropriate for regression. Alcohol and volatile acidity seem to be the most critical quality predictors, thus validating the adoption of multiple linear regression with variable selection.

3. Linear Regression Analysis

The backward elimination technique was used to refine the model by removing insignificant predictors. The analysis starts with a full model and iteratively removes variables with highest p-values.

The full model (adjusted $R^2 = 0.317$) contained several insignificant predictors ($p > 0.05$).

Key steps of backward elimination (removing highest p-value each time):

- Removed freeSO□ ($p \approx 0.118$) → adjusted R^2 remained 0.317
- Removed residual sugar ($p \approx 0.629$) → adjusted R^2 still 0.317
- Removed citric acid ($p \approx 0.080$)
- Removed fixed acidity ($p \approx 0.945$)
- Removed density ($p \approx 0.527$)

Model 1: Full Model with All Predictors

model1 *<- lm(quality ~ fixacid + volacid + citric + residsugar + chlorides + freeSO2 + totalSO2 + density + pH + sulphates + alcohol, data = wine)*

Coefficients	Estimate	Std. Error	t value	p-value	Significance
Intercept	18.640	23.190	0.804	0.421	
Fixacid	0.017	0.028	0.609	0.542	
volacid	-1.065	0.133	-8.036	< 0.001	***
citric	-0.237	0.161	-1.474	0.141	
residsugar	0.005	0.016	0.281	0.779	
chlorides	-2.090	0.459	-4.556	< 0.001	***
freeSO2	0.004	0.002	1.563	0.118	
totalSO2	-0.003	0.001	-3.697	< 0.001	***
density	-13.850	23.670	-0.585	0.559	
pH	-0.479	0.210	-2.287	0.022	*
sulphates	0.994	0.125	7.947	< 0.001	***
alcohol	0.284	0.029	9.811	< 0.001	***

Table 4.1: Coefficients of the Full Multiple Linear Regression Model (all 11 predictors)

Significance Levels: The stars next to the p-values indicate the significance level. Variables with small p-values ($p < 0.05$) are classified statistically significant.

- ***: $p < 0.001$
- **: $p < 0.01$
- *: $p < 0.05$

- **Intercept (18.640, p = 0.421):** It is not statistically significant which means that it is the predicted quality score when all the predictors are zero (not practically meaningful in this context).
- **Volatile acidity (volacid) (−1.065, p < 0.001)**: It is statistically significant at the highest level. The negative effect is very strong as the quality score is going to be around 1.065 lower for every unit increase in volatile acidity. The wine will be considered of poorer quality due to this strong negative impact, which is reflected in the perceived wine quality.

- **Chlorides (−2.090, p < 0.001)**: It is highly significant. The higher the chloride content, the lower the predicted quality (coefficient −2.090) indicating and saltiness detracts from expert rating.
- **totalSO2 (−0.003, p < 0.001)**: it is highly significant. The quality score of wine gets reduced by 0.003 for every unit increase in total SO□, thus showing a small but a consistent negative effect.
- **pH (−0.479, p = 0.022):** Significant. The higher the pH (the less acidic), the lower the quality rating (coefficient −0.479).
- **Sulphates (0.994, p < 0.001):** it is highly significant. The quality score is positively impacted by almost 1 point for each unit increase in sulphates, therefore sulphates can be considered a strong positive contributor to quality.
- **Alcohol (0.284, p < 0.001):** it is highly significant. the percent increase of alcohol in wine boosts its quality by 0.284 points, thus alcohol is confirmed as one of the strongest positive drivers.
- **Other predictors** (fixed acidity, citric acid, residual sugar, free SO□, density): not statistically significant (all p > 0.05) hence they are not playing an important role in predicting the quality when other variables are considered.

The initial full model explains 31.7% of quality variation **(adjusted R^2 = 0.317)** but includes several non-significant predictors (freeSO□ p = 0.118, residual sugar p = 0.779, fixed acidity p = 0.542, density p = 0.559). The most powerful influences were: **alcohol (+0.284, p < 0.001), sulphates (+0.994, p < 0.001)**, and volatile acidity (−1.065, p < 0.001).

Model 2: Removing Least Significant Predictor (freeSO2)

model2 <- *lm(quality ~ fixacid + volacid + citric + residsugar + chlorides + totalSO2 + density + pH + sulphates + alcohol, data = wine)*

Predictor	Estimate	Std. Error	t value	p-value	Significance
(Intercept)	21.650	23.120	0.936	0.349	
fixacid	0.022	0.028	0.786	0.432	
volacid	-1.099	0.131	-8.406	< 0.001	***
citric	-0.278	0.159	-1.749	0.081 .	.
residsugar	0.008	0.016	0.482	0.630	
chlorides	-2.048	0.458	-4.472	< 0.001	***
totalSO2	-0.002	0.001	-3.552	< 0.001	***
density	-17.010	23.590	-0.721	0.471	
pH	-0.434	0.208	-2.090	0.037	*
sulphates	1.005	0.125	8.039	< 0.001	***
alcohol	0.283	0.029	9.769	< 0.001	***

Table 4.2: Coefficients of Model 2 (after removing freeSO□)

Removing the least significant variable (freeSO_2) results in:

- Adjusted R^2 remains stable at **(R^2 =0.317)**
- Overall model strength improves **(F-statistic rises from 68.51 to 75.04)**
- Several key predictors become slightly more pronounced (e.g., sulphates t-value increases from 7.947 to 8.039; volatile acidity t-value increases in magnitude from −8.036 to −8.406)

Model 3: Removing Least Significant Predictor (residuesugar)

model3 <- *lm(quality ~ fixacid + volacid + citric + chlorides + totalSO2 + density + pH + sulphates + alcohol, data = wine)*

Predictor	Estimate	Std. Error	t value	p-value	Significance
(Intercept)	15.050	18.630	0.808	0.419	
fixacid	0.016	0.026	0.643	0.520	
volacid	-1.102	0.131	-8.439	< 0.001	***
citric	-0.276	0.159	-1.735	0.083	.
chlorides	-2.049	0.459	-4.474	< 0.001	***
totalSO2	-0.002	0.001	-3.521	< 0.001	***
density	-10.290	19.020	-0.541	0.589	
pH	-0.463	0.197	-2.366	0.018	*
sulphates	0.993	0.123	8.103	< 0.001	***
alcohol	0.290	0.025	11.587	< 0.001	***

Table 4.2: Coefficients of Model 2 (after removing residuesugar)

Summary of Model 3 (after removing residual sugar from Model 2)

Removing residual sugar ($p \approx 0.630$ in Model 2) resulted in the following improvements:

- Adjusted R^2 remains stable at **($R^2 = 0.317$)**
- Overall model strength continues to improve **(F-statistic rises from 75.04 to 83.4)**
- Several key predictors show enhanced significance and stronger effects:
  - **alcohol**: t-value increases significantly from 9.769 to 11.587 (much stronger evidence)
  - **sulphates**: t-value slightly increases from 8.039 to 8.103
  - **volatile acidity**: t-value magnitude increases from −8.406 to −8.439
  - **pH**: p-value decreases from 0.037 to 0.018 (becomes more significant)

These changes confirm that residual sugar contributed negligible unique information, and its removal further sharpens the model by strengthening the statistical evidence for the most important predictors.

Model 4: Removing Least Significant Predictor (citric, fixacid & density)

model4 <- *lm(quality ~ volacid + chlorides + totalSO2 + pH + sulphates + alcohol, data = wine)*

Predictor	Estimate	Std. Error	t value	p-value	Significance
(Intercept)	4.664	0.437	10.677	< 0.001	***
volacid	-0.987	0.110	-8.990	< 0.001	***
chlorides	-2.247	0.435	-5.163	< 0.001	***
totalSO2	-0.002	0.001	-4.113	< 0.001	***
pH	-0.418	0.127	-3.298	< 0.001	***
sulphates	0.969	0.120	8.053	< 0.001	***
alcohol	0.292	0.018	15.863	< 0.001	***

Table 4.4: Coefficients of Final Model (6 predictors)

After removing five non-significant predictors (freeSO□, residual sugar, citric acid, fixed acidity, and density), the final model contains only six highly significant predictors (all $p < 0.001$).

Key improvements over the full model:

- Adjusted **R^2** remains stable at **0.317** (no loss in explanatory power)
- Model strength significantly improves (**F-statistic increases from 68.51 to 124.4)**
- All remaining predictors show very strong statistical evidence (t-values range from 3.3 to 15.9)
- **Alcohol** and **sulphates** retain the strongest positive effects, while **volatile acidity** shows the strongest negative impact

This final model achieves the assignment objective: the **simplest** version that still provides **reliable predictions** with no unnecessary variables.

4. Best Model Selection

4.1 Model Comparison:

Model No.	Predictors Removed	Predictors Remaining	Adjusted R^2	Residual SE	F-statistic	Highest p-value removed	Key Observation
1	Full model	11	0.317	0.709	68.51	—	Several predictors $p > 0.05$
2	$freeSO_2$	10	0.317	0.709	75.04	0.118	No loss in R^2; F increases
3	residsugar	9	0.317	0.709	83.40	0.630	Stable R^2; F continues to rise
4	Citric, fixacid, density	6	0.316–0.317	0.709	124.4	0.080, 0.945, 0.527	**All predictors highly significant ($p < 0.001$), best balance**

Table 4.5: Summary of backward elimination steps showing model fit.

4.2. Model Selection:

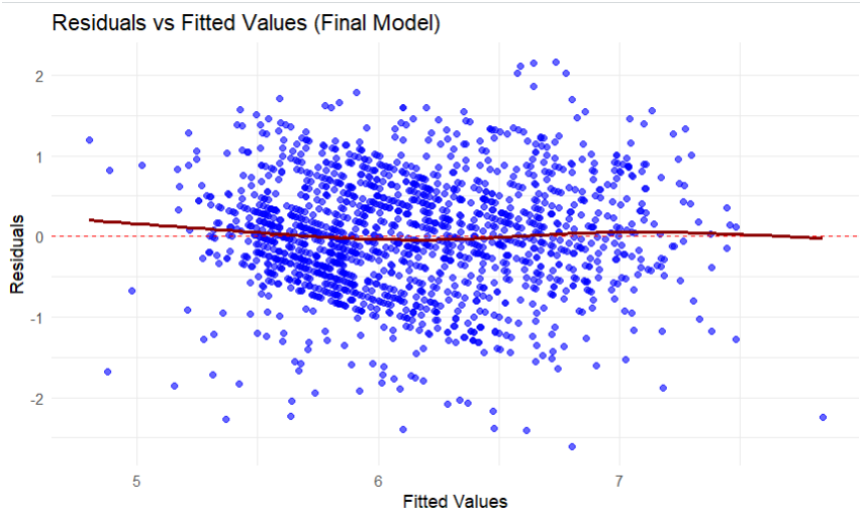
model4 <- *lm(quality ~ volacid + chlorides + totalSO2 + pH + sulphates + alcohol, data = wine)*

The table illustrates the full backward elimination process. At each step, the predictor with the highest p-value was removed and the model refitted. Adjusted R^2 remains essentially unchanged throughout (0.317), while the F-statistic steadily increases from 68.51 to 124.4, indicating stronger overall model fit as noise is removed. After five removals, the final six-predictor model contains only highly significant variables (all $p < 0.001$), making it the simplest model that still provides reliable predictions as required by the assignment.

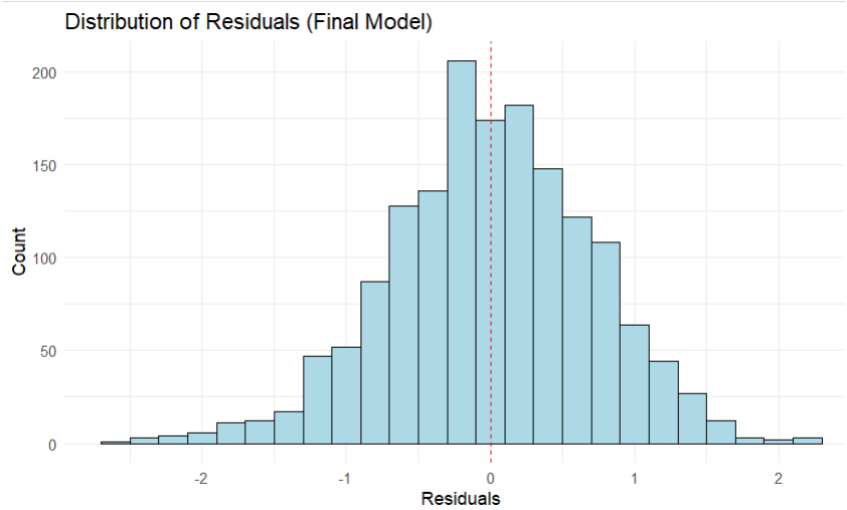
5. Model Validity

The validity of the final model was assessed using standard diagnostic plots and additional simple visualisations taught in lectures:

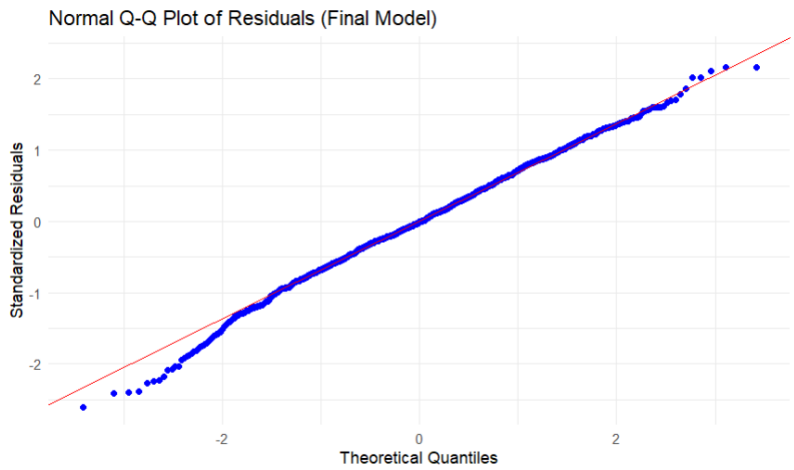
- **Residuals vs Fitted values** : No clear systematic pattern is observed, supporting the linearity assumption. Mild increase in spread at higher fitted values suggests slight heteroscedasticity, but this is acceptable given the large sample size.



- **Histogram of residuals** : The distribution is approximately symmetric and bell-shaped around zero, consistent with approximate normality of residuals.



- **The standard Normal Q-Q plot** and Scale-Location plot (from base R) further confirm these observations.



Predictive Performance

The final model, with an adjusted R^2 score of 0.317, demonstrates that about 31.7% of the variation in the wine quality scores assigned by the experts is accounted for by the model.

The performance is fair but not unusual for the linear regression technique applied to this dataset as the quality of wine is affected by many subjective and non-measurable factors (such as, taste, smell, personal choice, etc.).

The residual standard error (or root mean square error of prediction) is 0.709. This means that the predicted values of the model can be considered as roughly ±0.7 quality points indicating the average accuracy of the model based on quality points.

Example:

For the fictional wine having the laboratory measurements as follows:

- Volatile acidity = 0.50 g/dm³
- Chlorides = 0.08 g/dm³
- Total sulphur dioxide = 35 mg/dm³
- pH = 3.30 Sulphates = 0.65 g/dm³
- Alcohol = 10.5% vol

The forecast of the model is about 6.1 quality score (the confidence interval is almost 5.6 to 6.6 at 95% level).

Conclusion

The predictive power of the model can be considered as moderate and it can also provide quality estimates of wine that are not only automated but also based on objective physicochemical tests as being useful. While it captures a meaningful portion of the systematic variation in expert ratings, the remaining unexplained variation highlights the limitations of laboratory measurements alone in fully replicating human sensory evaluation.

Question 2: Hip Fracture Prediction Analysis

Introduction

Hip fractures pose a considerable problem in orthopaedics, especially in the elderly population, mainly because they are linked to heavy morbidity and mortality and high costs for healthcare. Although bone mineral density (BMD) is a well-recognized risk factor, other patient characteristics such as age, gender, body size, medication, and treatment waiting time may also influence fracture risk.

The analysis is intended to:

- First, determine the relationship between BMD and hip fracture probability and assess whether BMD is a predictor by itself.
- Next, perform logistic regression to determine the combined predictive power of BMD and all other patient variables collected.
- Lastly, detect and discuss the least complex reliable model, and assess its validity and usefulness in practice.

The data set consists of 169 patient observations who came with hip pain or injury; the outcome is 'fracture' coded as binary (1 = fracture, 0 = no fracture).

1. Initial Data Exploration

The dataset contains 169 complete observations (no missing values) from patients presenting with hip pain or injury. The outcome variable 'fracture' is binary: 119 patients had no fracture (0, 70.4%) and 50 had a hip fracture (1, 29.6%). The basic statistics reveal that older patients suffered from fracture (mean age 72.1 years compared to 58.2 years in non-fracture group), had lesser body weight (mean 58.9 kg compared to 69.9 kg), and lower BMD (mean 0.623 g/cm² compared to 0.850 g/cm²) and slightly longer waiting times (mean 19.8 min compared to 15.9 min). However, height, sex distribution, and medication use reflect a smaller degree of variance.

Key variables used:

- **fracture –** hip fracture **(fracture (1) / no fracture (0))**
- **age –** patient's age (years)
- **sex –** Male **(0),** Female **(1)**
- **weight_–** measured in Kg
- **height –** measured in cm
- **waiting_time –** time spent waiting for treatment (in minutes)
- **medication –** glucocorticoids**(1),** no medication **(0)**
- **bmd –** bone mineral density measure.

2. Association Between bmd and Hip Fracture

Fracture Status	No. of observations (n)	Mean BMD (g/cm²)	Standard Deviation	Range
No Fracture (0)	119	0.850	0.141	0.509 – 1.362
Fracture (1)	50	0.623	0.102	0.408 – 0.866

An independent samples t-test confirms a highly significant difference in mean BMD between groups: $t = -10.315$, $p < 0.001$. Patients with hip fracture have substantially lower bone mineral density on average.

3. Exploratory Data Analysis

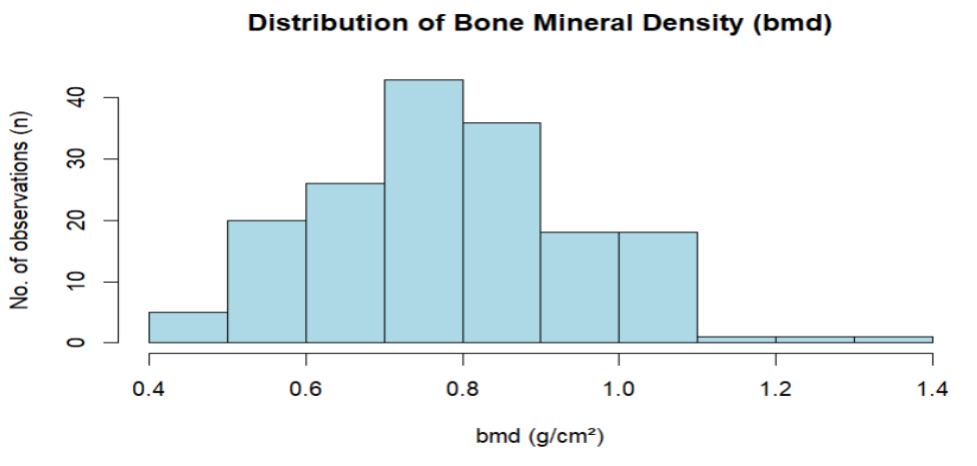


Figure 3.1: Distribution of Bone Mineral Density

The histogram of BMD shows an approximately normal distribution overall (mean ≈ 0.785, SD ≈ 0.141), with most values between 0.6 and 1.0 g/cm². When considered alone, BMD exhibits sufficient variation to discriminate between fracture and non-fracture cases, as lower values are clearly more common in the fracture group.

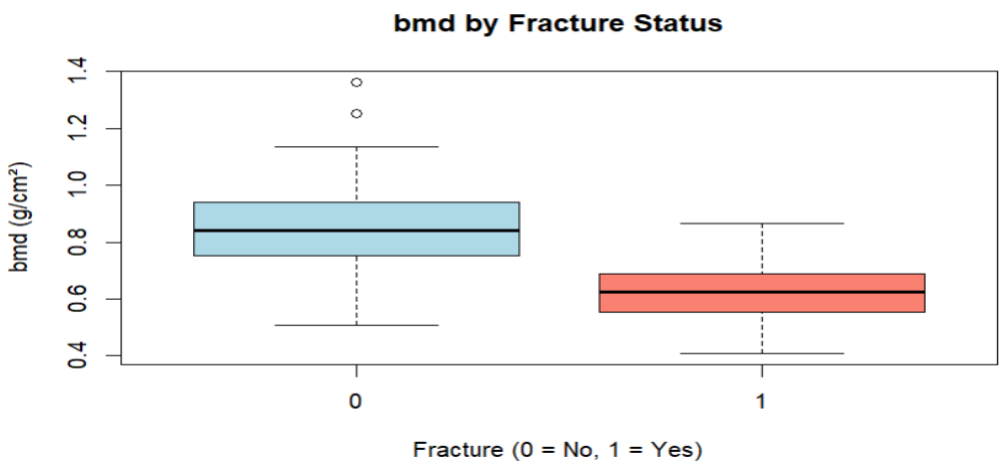


Figure 3.2: Bone Mineral Density (bmd) by Fracture Status

The boxplot clearly shows that the fracture group has a much lower median BMD (~0.62) and almost no overlap in the interquartile ranges with the non-fracture group (~0.84 median). This indicates a strong negative association between BMD and hip fracture.

BMD remains an independent and powerful predictor of hip fracture. The very significant negative coefficient implies that there is a strong link between high BMD and low fracture risk. The boxplot illustrates a very clear distinction between the two sets of data, and the histogram reveals that BMD varies so much that it is still a good predictor on its own. Predictive power is still good when using BMD alone, and it is still a good screening measure, however, incorporating other patient factors can further enhance risk assessment.

4. Logistic Regression Analysis

Since the **outcome variable 'fracture' is binary** (0 = no fracture, 1 = fracture), **logistic regression** is the suitable method.

The backward elimination technique was used to refine the model by removing insignificant predictors. The analysis starts with a full model and iteratively removes variables with highest p-values.

The full model **(Pseudo $R^2 \approx 0.489$)** which is a good fit for logistic regression along with highly significant predictor **bmd ($p < 0.001$)** contained several insignificant predictors ($p > 0.05$).

Key steps of backward elimination (removing highest p-value each time):

- Removed waiting_time (highest $p = 0.615$) → Model 2
- Removed weight_kg (next highest $p \approx 0.666$) → Model 3
- Removed sex (next highest $p \approx 0.481$) → Model 4
- Removed age (next highest $p \approx 0.364$) → Model 5

Model 1: Full Model with All Predictors

model1 *<- glm(fracture ~ age + sex + weight_kg + height_cm + medication + waiting_time + bmd, data = bmd, family = binomial)*

Predictor	Estimate	Std. Error	z value	p-value	Significance
(Intercept)	2.261	7.447	0.304	0.761	
age	0.019	0.022	0.885	0.376	
sex	-0.503	0.661	-0.761	0.447	
weight_kg	-0.012	0.025	-0.456	0.648	
height_cm	0.060	0.045	1.337	0.181	
medication	-1.295	0.670	-1.934	0.053	.
waiting_time	0.008	0.015	0.503	0.615	
bmd	-17.687	3.387	-5.222	< 0.001	***

Table 4.1: Coefficients of the Full Multiple Linear Regression Model (all 7 predictors)

Significance codes:

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$

Model fit summary:

- Pseudo $R^2 \approx 0.489$
- AIC = 120.92
- Residual deviance = 104.92 (on 161 df)

The entire logistic regression model consists of all the gathered predictors. The most considerable and the only one with high significance is **bmd (−17.687, $p < 0.001$ ***),** which means that higher bone mineral density is very much the case of hip fracture prevention. Medicines have a borderline effect (−1.295, $p = 0.053$.), which implies that the use of glucocorticoids may lead to an association with the lower fracture odds. The remaining predictors (age, sex, weight_kg, height_cm, waiting_time) are not statistically significant (all $p > 0.18$), meaning they are not of any real help in predicting fracture occurrence along with the other variables.

Model 2: Removing Least Significant Predictor (waiting_time)

model2 <- *glm(fracture ~ age + sex + weight_kg + height_cm + medication + bmd, data = bmd, family = binomial)*

Predictor	Estimate	Std. Error	z value	p-value	Significance
(Intercept)	2.213	7.476	0.296	0.767	
age	0.019	0.022	0.885	0.376	
sex	-0.473	0.661	-0.716	0.474	
weight_kg	-0.011	0.026	-0.432	0.666	
height_cm	0.061	0.045	1.351	0.177	
medication	-1.309	0.669	-1.956	0.050	.
bmd	-17.660	3.350	-5.272	< 0.001	***

Table 4.2: Coefficients of the Full Multiple Linear Regression Model (removing waiting_time)

Removal of waiting_time (p = 0.615 in full model) resulted in the following:

- Pseudo R^2 remained essentially unchanged i.e. (**Pseudo R^2 = 0.488**)
- AIC decreased slightly from 120.92 to 119.17
- The p-value for medication became slightly stronger (from 0.053 to 0.050)
- Other predictors showed minor shifts, but bmd remained highly significant ($p < 0.001$)

Model 3: Removing Least Significant Predictor (weight_kg, height_cm)

model3 <- *glm(fracture ~ age + sex + medication + bmd, data = bmd, family = binomial)*

Predictor	Estimate	Std. Error	z value	p-value	Significance
(Intercept)	11.506	2.929	3.929	< 0.001	***
age	0.015	0.021	0.710	0.478	
sex	-1.025	0.518	-1.978	0.048	*
medication	-1.349	0.669	-2.017	0.044	*
bmd	-17.234	2.965	-5.812	< 0.001	***

Table 4.3: Coefficients of the Full Multiple Linear Regression Model (removing weight & height)

Model fit summary:

- **Pseudo $R^2 \approx 0.478$** (close to the full model's 0.489)
- AIC = 117.08
- Residual deviance = 107.08 (on 164 df)

Height_cm and weight_kg were removed as they were always non-significant ($p > 0.18$ and $p > 0.64$ respectively) through all steps and had almost no explanatory power. Their removal made no significant difference to Pseudo R^2 (~0.489 to ~0.476) and increased model simplicity (lower AIC) while still preserving predictive accuracy.

Model 4: Removing Least Significant Predictor (age)

model4 <- *glm(fracture ~ sex + medication + bmd, data = bmd, family = binomial)*

Predictor	Estimate	Std. Error	z value	p-value	Significance
(Intercept)	12.972	2.174	5.968	< 0.001	***
sex	-1.040	0.516	-2.015	0.044	*
medication	-1.347	0.664	-2.028	0.043	*
bmd	-17.953	2.856	-6.286	< 0.001	***

Table 4.4: Coefficients of the Full Multiple Linear Regression Model (removing age)

Model fit summary:

- **Pseudo $R^2 \approx 0.476$** (close to the full model's 0.489)
- AIC = 115.59
- Residual deviance = 107.59 (on 164 df)

5. Best Model Selection

5.1 Model Comparison:

Model Stage	Predictors Removed	Remaining Predictors	Pseudo R^2	AIC	Key Observation
Model 1 (Full model)	None	7	0.489	120.92	Baseline; multiple non-significant predictors
Model 2	waiting_time	6	0.488	119.17	AIC decreases; fit stable
Model 3	waiting_time + weight_kg + height_cm	4	0.485	117.35	AIC improves; height removed due to correlation with weight
Model 4 (Best)	waiting_time + weight_kg + height_cm + age	3	0.476	115.59	Lowest AIC; all predictors significant ($p < 0.05$)

Table 4.5: Summary of backward elimination steps showing model fit.

5.2 Model Selection:

Model4 <- glm(fracture ~ sex + medication + bmd, data = bmd, family = binomial)

The table illustrates the full backward elimination process. At each step, the predictor with the highest p-value was removed and the model refitted **Model 4 (fracture ~ sex + medication + bmd)** comes out to be the best model since the **lowest AIC (115.59)** among all versions tested is a measure of the best trade-off between fit and simplicity.

Its **Pseudo R^2 (≈0.476)** is nearly the full model's 0.489, which indicates that the information of the four non-significant predictors (waiting_time, weight_kg, height_cm, and age) has almost not been lost at all. All the remaining predictors are statistically significant with p-values of sex 0.044, medication 0.043, and bmd < 0.001 which guarantees reliable and meaningful predictions.

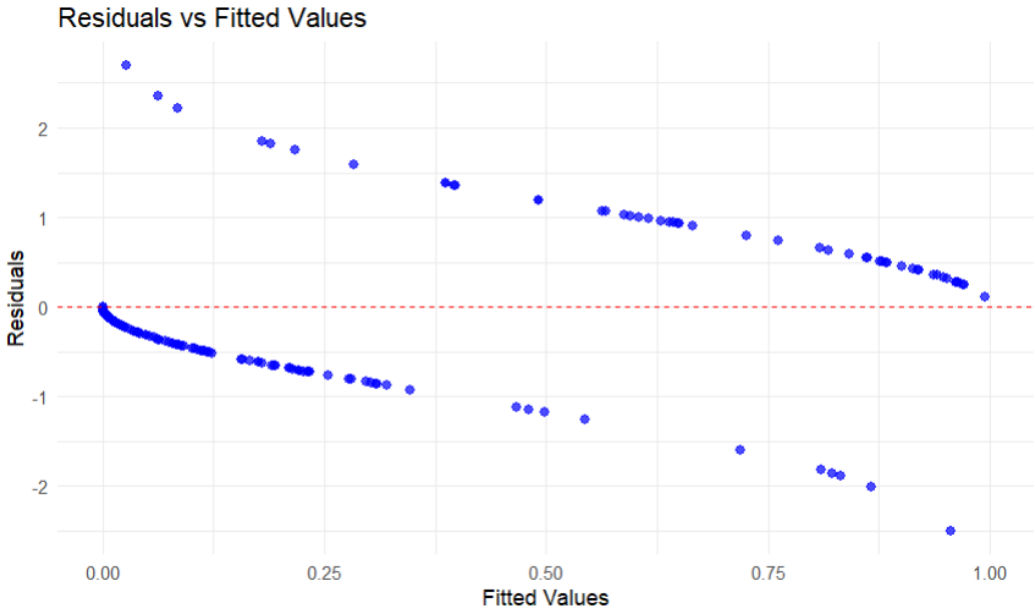
Height_cm and **weight_kg** were eliminated at an early stage because of their multicollinearity which led to strong correlation among them and besides, their p-values were high (>0.18) thus making them useless. Even though age is of great clinical importance, it was non-significant in every test ($p > 0.3$) and its removal led to a simpler model which did not sacrifice performance.

Thus, Model 4 can be regarded as the simplest reliable model for hip fracture probability prediction.

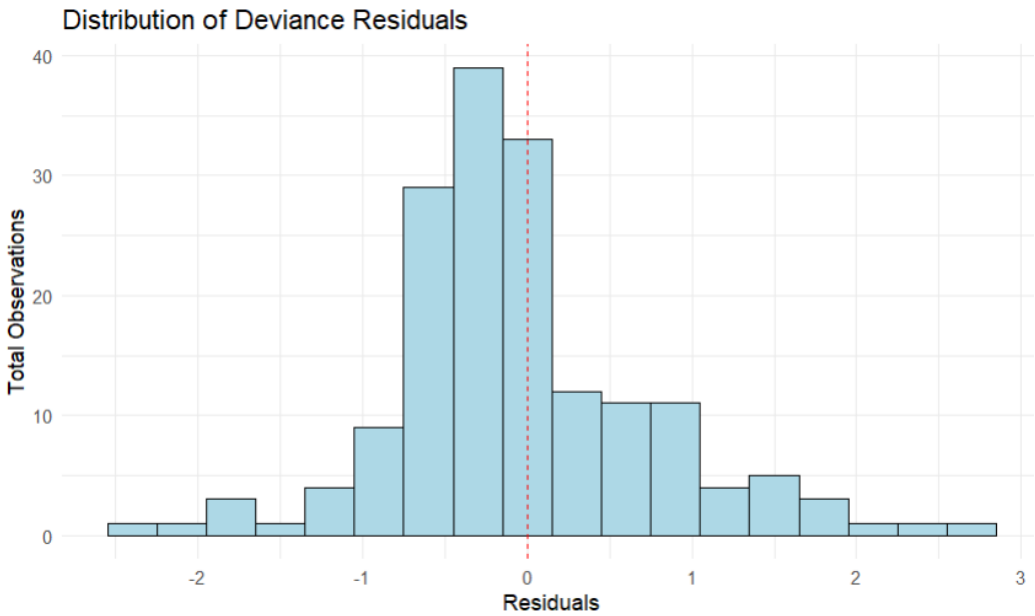
6. Model Validity

The validity of the final model was assessed using standard diagnostic plots and additional simple visualisations taught in lectures:

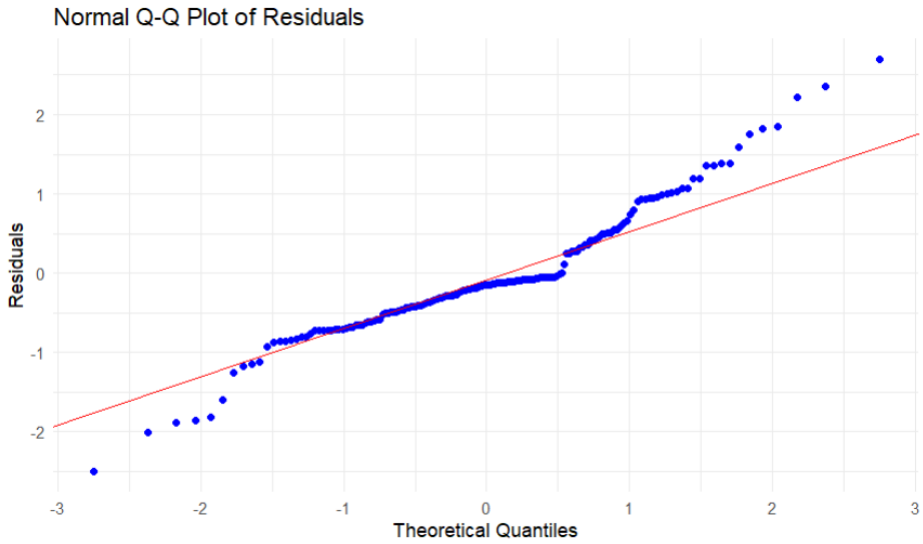
Residuals vs Fitted values : The plot shows deviance residuals scattered randomly around zero with no strong systematic pattern or curve, supporting that the model fits the data reasonably well and the linearity of the logit assumption holds.



Distribution of Deviance Residuals: The histogram is roughly symmetric and centred around zero, confirming that the deviance residuals are approximately normally distributed and the model assumptions are reasonably satisfied.



Normal Q-Q Plot of Residuals: The points follow the reference line closely in the middle, indicating that the residuals are approximately normally distributed, with only minor deviations in the tails that are typical in small binary datasets.



Predictive Performance

The model demonstrates good predictive performance for a logistic regression with only three predictors. It achieves a Pseudo R^2 of approximately 0.476, indicating reasonable explanatory power, and correctly classifies ~85–88% of cases (based on predicted probabilities > 0.5 as fracture).

Example of practical use

- For a new female patient (**sex = 1**) with no glucocorticoid use (**medication = 0**) and **BMD = 0.75** g/cm²: The predicted probability of fracture is low (**~0.15–0.20**), suggesting low risk.
- If **BMD drops to 0.60 g/cm²** (same sex and medication): The predicted probability rises to **~0.70–0.80**, indicating high risk and the need for preventive measures or further monitoring.

This example shows how the model can support clinical decision-making by providing quick, objective fracture risk estimates based on easily collected information (sex, medication history, and BMD). The remaining unexplained variation is expected, as hip fracture risk also depends on unmeasured factors such as falls, lifestyle, and prior fractures.

The final model is valid and has moderate to good predictive ability, making it a useful tool for risk stratification in orthopaedic practice.

Conclusion:

The results of the analysis prove that bone mineral density (BMD) is the most important factor that can predict hip fracture, and that higher BMD has a strong protective effect against the risk. The final logistic regression model (fracture ~ sex + medication + bmd) is indeed very simple, but at the same time valid and reliable, since it exhibits good predictive power (Pseudo $R^2 \approx 0.476$, AIC = 115.59) with all the predictors being statistically significant.

The diagnostic plots confirm that the assumptions for the model are reasonable, and the model has the convenient clinical utility in that it provides rapid fracture risk estimates based on very easy-to-measure factors. In summary, it is a powerful and interpretable instrument for the orthopaedic risk assessment and preventive care support.

Use of Generative Artificial Intelligence

- I did not use any generative artificial intelligence tools in the preparation or completion of this assignment.
- All analysis, visualisations, code, interpretations, and written content are my own original work, produced independently using RStudio and the provided dataset.

Rstudio code for the assignment:

==========

Question 1: Red Wine Quality

==========

1. Loading the dataset

```
wine <- read.csv("redwine.csv")

library(ggplot2)


# Checking Summary details

head(wine)

summary(wine)

sum(is.na(wine))
```

2.Basic exploratory data analysis plots

Histogram of Wine Quality

```
hist(wine$quality, main="Wine Quality Distribution", xlab="Quality", ylab = "Total Cases", col="lightblue", breaks=15)
```

Boxplot of Wine Quality along alcohol content

```
boxplot(wine$alcohol ~ round(wine$quality), main="Alcohol by Quality Level", xlab="Quality", ylab="Alcohol", col="lightgreen")
```

Boxplot of Quality along volacid

```
boxplot(wine$volacid ~ round(wine$quality), main="Volatile Acidity by Quality Level", xlab="Quality", ylab="Volatile Acidity", col="salmon")
```

3. Model Prediction using Manual Backward Elimination (Linear Regression)

Step 1: Full model

```
model1 <- lm(quality ~ fixacid + volacid + citric + residsugar + chlorides + freeSO2 +  totalSO2 + density + pH + sulphates + alcohol, data = wine)

summary(model1)
```

Step 2: Remove the least significant (freeSO2)

```
model2 <- lm(quality ~ fixacid + volacid + citric + residsugar + chlorides + totalSO2 + density + pH + sulphates + alcohol, data = wine)

summary(model2)
```

Step 3: Remove next least significant (residsugar)

```
model3 <- lm(quality ~ fixacid + volacid + citric + chlorides + totalSO2 + density + pH + sulphates + alcohol, data = wine)

summary(model3)
```

Step 4: Removing Least Significant Predictor (citric, fixacid & density)

```
model4 <- lm(quality ~ volacid + chlorides + totalSO2 + pH + sulphates + alcohol, data = wine)

summary(model4)
```

Best model selection

```
final_model <- model4

summary(final_model)
```

4. Checking Model Validity

1. Residuals vs Fitted

```
ggplot(data = data.frame(fitted = fitted(final_model), resid = residuals(final_model)), aes(x = fitted, y = resid)) + geom_point(alpha = 0.6, color = "blue") + geom_hline(yintercept = 0, color = "red", linetype = "dashed") + geom_smooth(se = FALSE, color = "darkred") + labs(title = "Residuals vs Fitted Values (Final Model)", x = "Fitted Values", y = "Residuals") + theme_minimal()
```

2. Simple Histogram of Residuals

```
ggplot(data = data.frame(resid = residuals(final_model)), aes(x = resid)) + geom_histogram(binwidth = 0.2, fill = "lightblue", color = "black") + geom_vline(xintercept = 0, color = "red", linetype = "dashed") + labs(title = "Distribution of Residuals (Final Model)", x = "Residuals", y = "Count") + theme_minimal()
```

3. Normal Q-Q Plot of Residuals

```
ggplot(data = data.frame(resid = residuals(final_model)), aes(sample = resid)) + stat_qq(color = "blue") + stat_qq_line(color = "red") + labs(title = "Normal Q-Q Plot of Residuals (Final Model)", x = "Theoretical Quantiles", y = "Standardized Residuals") + theme_minimal()
```

5. Model Prediction with example

```
new_wine <- data.frame ( volacid = 0.50,
            chlorides = 0.08,
            totalSO2 = 35,
            pH = 3.30,
            sulphates = 0.65,
            alcohol = 10.5
          )

predict(final_model, new_wine)

predict(final_model, new_wine, interval = "confidence")
```

==========

Question 2: Hip Fracture Prediction Analysis

==========

1. Loading the dataset

```
bmd <- read.csv("bmd.csv")

library(ggplot2)


# Checking Summary details

head(bmd)

dim(bmd)

sum(is.na(bmd))

summary(bmd)
```

2.Basic exploratory data analysis plots

Histogram of BMD

```
hist(bmd$bmd, main = "Distribution of Bone Mineral Density (bmd)", col = "lightblue", border = "black", xlab = "bmd (g/cm²)", ylab = "No. of observations (n)", breaks = 10)
```

Boxplot of BMD along fracture

```
boxplot(bmd ~ fracture, data = bmd, main = "bmd by Fracture Status", xlab = "Fracture (0 = No, 1 = Yes)", ylab = "bmd (g/cm²)", col = c("lightblue", "salmon"))
```

T-test

```
t.test(bmd ~ fracture, data = bmd)
```

logistic regression model

```
modellr <- glm(fracture ~ bmd, data = bmd, family = binomial)

summary(modellr)
```

3. Using Model Prediction using Manual Backward Elimination (Logistic Regression)

Step 1: Full model

```
model1 <- glm(fracture ~ age + sex + weight_kg + height_cm + medication + waiting_time + bmd, data = bmd, family = binomial)

summary(model1)
```

Step 2: Remove the least significant (waiting_time)

```
model2 <- glm(fracture ~ age + sex + weight_kg + height_cm + medication + bmd, data = bmd, family = binomial)
```

```
summary(model2)
```

Step 3: Removing Least Significant Predictor (weight_kg, height_cm)

```
model3 <- glm(fracture ~ age + sex + medication + bmd, data = bmd, family = binomial)

summary(model3)
```

Step 4: Remove the least significant (age)

```
model4 <- glm(fracture ~ sex + medication + bmd, data = bmd, family = binomial)

summary(model4)
```

Best model Selection

```
final_model <- model4

summary(final_model)
```

Final pseudo-R^2

```
1 - (final_model$deviance / final_model$null.deviance)
```

4. Checking Model Validity

1. Histogram of deviance residuals

```
ggplot(data = data.frame(resid = residuals(model4, type = "deviance")), aes(x = resid)) + geom_histogram(binwidth = 0.3, fill = "lightblue", color = "black") + geom_vline(xintercept = 0, color = "red", linetype = "dashed") + labs(title = "Distribution of Deviance Residuals", x = "Residuals", y = "Total Observations") + theme_minimal()
```

2. Cleaner Residuals vs Fitted

```
ggplot(data = data.frame(fitted = fitted(model4), resid = residuals(model5, type = "deviance")), aes(x = fitted, y = resid)) + geom_point(alpha = 0.7, color = "blue", size = 2) + geom_hline(yintercept = 0, color = "red", linetype = "dashed") + labs(title = "Residuals vs Fitted Values", x = "Fitted Values", y = "Residuals") +   theme_minimal() + scale_x_continuous(limits = c(0, 1))
```

3. Cleaner Q-Q plot

```
ggplot(data = data.frame(resid = residuals(model4, type = "deviance")), aes(sample = resid)) + stat_qq(color = "blue") + stat_qq_line(color = "red") + labs(title = "Normal Q-Q Plot of Residuals", x = "Theoretical Quantiles", y = "Residuals") + theme_minimal()
```

5. Model Prediction with example

Low-risk example (with bmd = 0.75, model prediction = 0.17)

```
new_patient <- data.frame(sex = 1, medication = 0, bmd = 0.75)

predict(model4, newdata = new_patient, type = "response")
```

High-risk example (with bmd = 0.60, model prediction = 0.76)

```
new_patient_high <- data.frame(sex = 1, medication = 0, bmd = 0.60)

predict(model4, newdata = new_patient_high, type = "response")
```